

Database design

1) ENUM Types

Membership Type: Basic, Student, Premium

Book Copy Status: Available, Borrowed, Damaged

Borrow Status: Borrowed, Returned, Overdue, Lost

2) Tables Overview

MembershipTypes (membership limits),

Members (member details),

BookCategories (book categories),

Authors (author details),

Books (book information),

BookAuthors (book-author mapping),

BookCopies (physical copies),

BorrowTransactions (borrow records),

Fines (fine details),

FinePayments (fine payment history)

3) Table Design

1) MembershipTypes(MembershipTypeId, TypeName, MaxBooks, MaxBorrowDays)

Relationship: One membership type -> Many members

2) Members(MemberId, FullName, Email, Phone, Address, MembershipTypeId, IsActive)

Relationship: One member -> Many borrow transactions

3) BookCategories(CategoryId, CategoryName, Description)

Relationship: One BookCategory -> Many Books

4) Authors(AuthorId, AuthorName)

Relationship: Many Authors <-> Many Books

5) Books(BookId, Title, ISBN, PublishedYear, CategoryId)

Relationship: One Book -> Many BookCopies

Relationship: Many Books <-> Many Authors

6) BookAuthors(BookId, AuthorId)

Relationship: Many Books <-> Many Authors

7) BookCopies(CopyId, BookId, CopyCode, Status, ShelfLocation, IsAvailable, IsDamaged)

Relationship: One Book -> Many BookCopies

8) BorrowTransactions(BorrowId, MemberId, CopyId, BorrowDate, DueDate, ReturnDate, Status)

Relationship: One Member -> Many BorrowTransactions

Relationship: One BookCopy -> Many BorrowTransactions

9) Fines(FineId, BorrowId, MemberId, FineAmount, PaidAmount, PendingAmount, IsPaid)

Relationship: One BorrowTransaction -> One Fine

10) FinePayments(PaymentId, FineId, AmountPaid, PaymentDate)

Relationship: One Fine -> Many FinePayments